2026 Sessions

Min, B.Y., Wake B.E., Jain, R., An Investigation of CFD Predictions for Rotor Hover Performance Near Stall Conditions," AIAA 2026-1509, 64th AIAA Aerospace Sciences Meeting, Orlando, FL, January 12 – 16, 2025.

Accurate rotor stall prediction is critical to determine rotor's maximum performance and stall boundary. While Computational Fluid Dynamics (CFD) has shown significant improvement in accuracy over the past several decades, accurate prediction of separation and stall remains a challenge. Often, suspicious abrupt stall is observed from the CFD simulations at high thrust conditions. In this study, the issue is demonstrated with two publicly available rotors: Hover Validation and Acoustic Baseline (HVAB) rotor and XV-15 rotor. Furthermore, fictitious Simplified Blade (SB) models were used to examine tip design (anhedral and different airfoils) impact on stall behavior. Also, a static VR-12 airfoil model was used to examine various numerical schemes for stall prediction. The blade tip anhedral design delayed stall by altering blade-vortex interaction strength. The rotor stall prediction followed trend of the tip airfoil stall prediction. For the XV-15 rotor tested outdoor, it was found that headwind situation could have favorable impact on stall. Among tested turbulence models, Menter's Shear Stress Transport (SST) model and Spalart-Allmaras (SA) Delayed Detached Eddy Simulation (DDES) model showed premature stall. The standard Menter baseline two-equation model (BSL) showed delayed stall from the VR-12 airfoil study, and this behavior was maintained for rotor simulations, making improved correlation with test data where previous models failed with sudden stall. Current study indicated further attention to the turbulence model improvement is necessary for accurate stall prediction.

Abras, J., Lakshminarayan, V., Mestreau, E., Dey, S., Hariharan, N., "Enhanced Engineering Workflows in the HPCMP CREATE™ Ecosystem Demonstrated on the SUI Endurance UAV," AIAA 2026-1510, 64th AIAA Aerospace Sciences Meeting, Orlando, FL, January 12 – 16, 2025.

Advancements have been made in recent years focusing on computational solver speed, but while this is a significant factor in the time needed for the full solution, it is not the only area where time savings must be achieved. Enhancing the full engineering workflow is necessary for computational simulations to contribute to engineering processes in a timely manner. The full workflow encompasses the moment the engineer receives the geometry description and conditions until the moment the results are distilled and reported. The CREATE ecosystem has methods to support the full workflow and these methods are continuously improving. In this effort recent advances are demonstrated across the board starting with a new shrink-wrapping capability in HPCMP CREATE™-FT Capstone that eliminates the time needed to cleanup manufacturer provided geometry files. An auto-mesh feature is introduced that generates the computational grids by calling Capstone from within HPCMP CREATE™-AV Helios off-loading this time-consuming step to cluster execution and allowing for more rapid grid studies. These meshes are employed to execute more efficient GPU based simulations using Helios for both high-fidelity and mid-fidelity methods. These time saving capabilities are demonstrated for the realistic SUI Endurance quadcopter configuration and full vehicle performance results are compared to available test data.

Sheng, C., "Numerical Simulations of an Installed Rotor in Prescribed and Free Flight Maneuvers'" AIAA 2026-1511, 64th AIAA Aerospace Sciences Meeting, Orlando, FL, January 12 – 16, 2025.

This study develops a dynamic mesh framework for simulating rotorcraft in prescribed and freeflight maneuvers by coupling high-fidelity computational fluid dynamics (CFD) with a sixdegree-of-freedom (6-DOF) flight dynamics model. The CFD solver uses a hybrid RANS–LES approach on unstructured meshes to capture complex flow behaviors across various regimes. Aerodynamic loads computed by the CFD solver

are transferred to the 6-DOF model to update the vehicle's position and orientation. A dynamic mesh framework is developed to investigate the vehicle's flight dynamics. The current work provides a basis for high-fidelity rotorcraft dynamics studies, with further extensions to incorporate advanced flight-control models for assessing vehicle response and operational safety.

Schaefer, J.A., Abras, J.N., Narducci, R.P., Wissink, A.M., Jain, R.K., "Analysis of HVAB Rotor Hover Performance with Uncertainty in CFD Meshing Strategies," AIAA 2026-1512, 64th AIAA Aerospace Sciences Meeting, Orlando, FL, January 12 – 16, 2025.

Over the past few decades, advances in computing hardware capabilities have enabled increasingly higher fidelity Computational Fluid Dynamics (CFD) simulations to be performed for a variety of aerospace applications. One area of particular importance to the rotorcraft community is the prediction of hover performance. Although the computational cost of hover simulations is tractable – especially for the steady-state calculations considered in this work – ensuring their accuracy remains a challenge. There are many sources of uncertainty that could affect the results of hover simulation predictions, but this work aims to focus solely on the importance of discretization error due to the use of finite resolution meshes in steady-state NASA OVERFLOW simulations. Specifically, several common choices for best-practice gridding parameters are considered to be uncertain, and the effects of this uncertainty are quantified using a combination of automated parametric grid generation and surrogate modeling techniques. The Hover Validation and Acoustic Baseline (HVAB) rotor was chosen as the geometry due to its frequent usage in previous AIAA Hover Prediction Workshop studies. Surrogate modeling techniques include the Non-Intrusive Polynomial Chaos, Gaussian Process Regression, and two Deep Neural Network approaches due to their common usage for CFD and other Uncertainty Quantification (UQ) applications. Among the six uncertain gridding parameters considered, the first spacing off the wall and leading edge spacing were found to have the largest impact on the uncertainty in HVAB hover performance. If future studies aim to reduce discretization error in hover simulation predictions, these two parameters should be a primary focus.

Nelson, C., Ku, M., Yanagita, Y., Sankar, L., "Preliminary Application of Inverse Design to the HVAB Rotor Tip," AIAA 2026-1513, 64th AIAA Aerospace Sciences Meeting, Orlando, FL, January 12 – 16, 2025.

An inverse design procedure was applied to the HVAB rotor to develop a modified tip geometry which would provide improved hover performance. The design was tested using the GT-Hybrid tool over a range of conditions and was found to consistently provide 2-3% improvement in the figure of merit. As a further check on the results, the Simcenter STAR-CCM+ multiphysics package was used to model the 10 degrees collective condition. Preliminary results show a performance improvement from the modified tip geometry that is consistent with that predicted by GT-Hybrid.